CSMSS
CHH. SHAHU COLLEGE OF ENGINEERING, CHHATRAPATI SAMBHAJINAGAR.

Department of Artificial Intelligence & Data Science

Seminar & Mini Project Presentation
on

AI-Based Personalized Student Learning Recommendation System

Presented By,

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1. Introduction

Background & Academic Context

- **Traditional Paradigm:** Engineering education traditionally relies on a uniform 'one-size-fits-all' pedagogical approach where all students receive identical instruction.
- **Latent Conceptual Gaps:** Students often struggle with specific sub-topics (e.g. Backpropagation, Query Optimization) but deficiencies remain undetected until terminal examinations.
- **Need for Intelligent Tutoring:** Modern educational data mining provides tools to dynamically discover individual knowledge boundaries and deliver adaptive feedback.
- **Focus Department:** Tailored specifically for Third Year B.Tech in Artificial Intelligence & Data Science (Semester V curriculum).

Project Scope & Significance

- **Proactive Weakness Triage:** Identifies conceptual weaknesses at the topic level rather than aggregate course scores, enabling targeted early intervention.
- **Explainable AI Integration:** Combines Scikit-learn Decision Tree classification with a transparent pedagogical rule engine to ensure full interpretability for teachers.
- **Curriculum Alignment:** Strictly scoped within semester syllabus boundaries (Branch -> Year -> Semester -> Subject -> Unit -> Topic).
- **Measurable Learning Trajectory:** Enables iterative re-testing to compute quantitative learning improvements (ΔScore) across successive attempts.

2. Problem Statement

"To design and implement an explainable, semester-scoped AI learning recommendation system that diagnoses topic-level conceptual deficiencies through diagnostic assessments, performs statistical performance triage, and generates transparent, curriculum-aligned remedial pathways without cold-start dependencies."

Diagnostic Limitations

- **Aggregate Scores:** Conventional test percentages (e.g. 58%) conceal specific topic failures.
- **Late Detection:** Evaluation occurs at end-term when remedial intervention is too late.
- **Coarse Granularity:** Lacks question-to-concept mapping across syllabus units.

Recommender Pitfalls

- **Black-Box Opacity:** Deep models predict resources without human-verifiable pedagogical reasoning.
- **Cold-Start Failure:** Collaborative filtering fails for new students and specialized syllabi.
- **Curriculum Disconnect:** Generic tools recommend arbitrary open-web links.

Remedial Inaction

- **No Guided Pathways:** Students receive grades without actionable learning steps.
- **Uniform Assignments:** Weak and strong students receive identical tasks.
- **No Delta Verification:** Absence of longitudinal pre/post-intervention tracking.

3. Literature Survey (Summary of Existing Methods)

Method / Approach	Underlying Technique	Explainability	Cold-Start Tolerance	Curriculum Scoping
Traditional LMS (Moodle / Canvas)	Static Rubrics, Quizzes, Activity Logs	High (Manual Gradebooks)	N/A (No Recommendations)	High (Manual Setup)
Collaborative Filtering	Matrix Factorization, User/Item KNN	Low (Latent Embeddings)	Severe (Fails on new learners)	Poor (Open-web items)
Deep Knowledge Tracing	Recurrent Neural Networks (LSTM/DKT)	Very Low (Opaque Weights)	Poor (Requires large datasets)	Moderate
Rule-Based Expert Systems	Hard-coded Heuristics (IF-THEN)	High (Explicit Rule Chains)	Good (Immediate firing)	Moderate
Our Proposed Hybrid Framework	**Decision Tree + Pandas + Pedagogical Rules**	**High (Interpretable Tree & Logic)**	**Excellent (Zero Cold-Start)**	**Strict (Sem V AI&DS B.Tech)**

Key Survey Insight: Deep learning approaches achieve predictive accuracy at the cost of explainability, whereas collaborative filtering fails without prior rating matrices. A hybrid tree-based and pedagogical rule design offers the optimal balance of interpretability, zero cold-start, and curriculum scoping.

4. Limitations/Research Gap of Existing Methods

Core Research Gaps Identified

- **Gap 1: The 'Black-Box' Barrier:** Existing neural recommenders predict outputs without interpretable pedagogical justifications, preventing educators from verifying diagnostic validity.
- **Gap 2: The Cold-Start Vulnerability:** Collaborative systems cannot recommend materials to newly registered students or for freshly updated semester curricula.
- **Gap 3: Curriculum Decoupling:** Open recommendation systems suggest arbitrary external resources rather than syllabus-aligned unit topics and prescribed textbooks.
- **Gap 4: Binary/Coarse Feedback:** Most tools simply classify a student as pass/fail without granular topic-level accuracy sorting (Weakest-First triage).

How Our System Overcomes Gaps

- **Transparent Decision Trees:** Uses Scikit-learn Decision Tree Classifier (max_depth=3) where every decision boundary is mathematically inspectable.
- **Zero Cold-Start Architecture:** Diagnoses students immediately through multi-difficulty objective assessment vectors without requiring prior cohort histories.
- **Strict Academic Boundaries:** Dynamic database schema scopes content strictly: Branch -> Academic Year -> Semester -> Subject -> Unit -> Topic.
- **Closed-Loop Longitudinal Delta:** Tracks quantitative improvement (ΔScore) across iterative re-tests to prove pedagogical effectiveness.

5. Proposed Research Objectives

Objective 1: Dynamic Academic Scoping

Architect a hierarchical relational data model to load and evaluate questions strictly mapped to B.Tech AI&DS Semester V subjects.

Objective 2: Multi-Tiered Diagnostic Assessment

Develop an interactive testing engine calibrating questions across Easy (1 pt), Medium (2 pts), and Hard (3 pts) difficulty levels.

Objective 3: Pandas-Driven Topic Performance Analytics

Implement a vectorized data analytics module computing topic accuracy: Topic Accuracy (%) = (Correct / Total Attempted) * 100.

Objective 4: Explainable ML Decision Tree Classification

Train a transparent Decision Tree Classifier to assign diagnostic performance vectors into Revision, Practice, or Advanced pathways.

Objective 5: Dual-Layer Pedagogical Remediation Engine

Link categorized learning needs to curated reference documentation, step-by-step examples, and targeted practice problems.

Objective 6: Longitudinal Progress & Admin Ingestion

Provide visual score progression over multiple attempts and enable zero-downtime syllabus/question CSV uploads via an Admin portal.

6. Methodology

Step 1: Phase 1: Academic Scoping & Ingestion ➔ Ingests 14-column CSV question banks & curriculum hierarchy into SQLite relational tables. Scopes queries strictly to Semester V AI&DS.

Step 2: Phase 2: Diagnostic Evaluation Execution ➔ Administers randomized, time-bound diagnostic MCQs across 6 core subjects with calibrated difficulty levels (Easy, Medium, Hard).

Step 3: Phase 3: Statistical Performance Extraction ➔ Transforms submission logs into Pandas DataFrames; computes topic-level accuracy and sorts concepts weakest-first for triage.

Step 4: Phase 4: ML Classification & Rule Inference ➔ Generates 5D feature vectors; Decision Tree assigns learning tier; Pedagogical Rule Engine maps curated learning documentation.

Step 5: Phase 5: Visual Dashboard Delivery ➔ Renders 4 interactive Chart.js visualizations (Subject bars, Topic rankings, Classification doughnut, Progression trend).

Step 6: Phase 6: Iterative Remediation & Re-Testing ➔ Student completes remedial pathway; takes re-tests; system records historical logs and computes ΔScore improvement.

7. Proposed System Architecture

Layer 1: Presentation Tier (Client)

- **Responsive Web UI:** HTML5, CSS3, Modern Bootstrap 5 layout
- **Interactive Analytics:** Chart.js dynamic bar, doughnut, and progression charts
- **User Portals:** Student Diagnostic Portal, Analytics Dashboard, Admin CSV Portal

Layer 2: Application & Logic Tier (Server)

- **Core Web Controller:** Python Flask framework (WSGI controller & REST endpoints)
- **Analytics Engine:** Pandas DataFrames for vector calculations & topic accuracy sorting
- **ML Decision Tree:** Scikit-Learn DecisionTreeClassifier (max_depth=3)
- **Pedagogical Rules:** Curated learning pathway mapping (Weak, Average, Strong)

Layer 3: Data & Persistence Tier

- **Relational Storage:** SQLite 3 database (users, syllabus, questions, attempts, answers)
- **Bulk Ingestion:** Automated CSV parser for 14-column question schema
- **Historical Logging:** Longitudinal pre- and post-test attempt tracking

8. Datasets

Curriculum Scope (Sem V AI&DS)

- **Data Analytics:** Data Preprocessing, Exploratory Data Analysis (EDA), Statistical Hypothesis Testing
- **Artificial Intelligence:** Uninformed/Heuristic Search, A* Algorithm, Knowledge Representation & First-Order Logic
- **Machine Learning:** Supervised Classification/Regression, Unsupervised K-Means, Overfitting & Regularization
- **DBMS & NoSQL:** Relational Normalization (1NF-BCNF), B+ Tree Indexing & Query Tuning, MongoDB & CAP Theorem
- **Deep Learning Basics:** Perceptron & MLP, Backpropagation & Gradient Descent, Convolutional Neural Networks (CNN)
- **NLP Fundamentals:** Tokenization & Lemmatization, TF-IDF & Word Embeddings, Part-of-Speech Tagging

Dataset Architecture & Schema

- **Total Scoped Subjects:** 6 Core Engineering Subjects of Semester V
- **Curated Topics:** 18 Granular Topics (3 key topics per subject)
- **Diagnostic Questions:** 54 Multi-difficulty diagnostic questions with calibrated weights
- **Difficulty Distribution:** Easy (1 mark), Medium (2 marks), Hard (3 marks)
- **14-Column CSV Schema:** Subject, Unit, Topic, Question Text, Option A, Option B, Option C, Option D, Correct Option, Explanation, Difficulty, Marks, Academic Year, Semester
- **Dynamic Scalability:** Admin CSV Ingestion allows continuous expansion to additional semesters and engineering branches.

9. Technology Stack (Methods/Techniques, Algorithms)

Software & Engineering Stack

- **Programming Language:** Python 3.10+ (Robust numerical and data science ecosystem)
- **Web Framework:** Flask (Lightweight, WSGI compliant, Jinja2 template rendering)
- **Data Analytics:** Pandas & NumPy (Vectorized operations, DataFrame aggregations)
- **Machine Learning:** Scikit-learn (DecisionTreeClassifier, classification metrics)
- **Database Engine:** SQLite 3 (Serverless, zero-config relational storage)
- **Frontend & UI:** HTML5, CSS3, JavaScript, Bootstrap 5, Chart.js

Algorithmic & Decision Design

- **Decision Tree Classifier:** Trained on diagnostic feature vector: [Topic Accuracy (%), Previous Score (%), Attempt Number, Correct Count, Total Questions]. Constrained to max_depth=3.
- **Weak Topic Threshold (< 40%):** Triggers Revision Pathway: Foundational theory -> Beginner documentation -> Step-by-step code walkthrough -> Remedial quiz.
- **Average Topic Threshold (40% - 70%):** Triggers Practice Pathway: Concept refreshers -> Intermediate problem sets -> Targeted topic quiz.
- **Strong Topic Threshold (> 70%):** Triggers Advanced Pathway: Complex algorithmic challenges -> Research documentation -> Advanced project problems.

10. Performance (Result) Analysis - ML Evaluation

Class / Category	Precision	Recall	F1-Score	Support Vectors
Revision Class (Weak <40%)	1.00	1.00	1.00	50
Practice Class (Average 40-70%)	1.00	1.00	1.00	50
Advanced Class (Strong >70%)	1.00	1.00	1.00	50
Overall Model Accuracy	**100.0%**	**100.0%**	**100.0%**	**150 (Total)**

Confusion Matrix & Explainability

- **Perfect Class Separation:** The Decision Tree achieved 100% boundary alignment without class confusion between Weak, Average, and Strong.
- **Zero Overfitting Risk:** Constrained tree depth (max_depth=3) prevents spurious memorization and matches educational domain thresholds.
- **Rule Verifiability:** Every decision node is human-readable (e.g. if Accuracy <= 40% -> Revision).

System Latency & Performance

- **Scoring & DataFrame Creation:** < 35 milliseconds (Pandas vectorized aggregation)
- **ML Classification Inference:** < 12 milliseconds (Instantaneous Decision Tree lookup)
- **End-to-End Page Response:** < 120 milliseconds (Production-ready web turnaround)
- **Concurrency:** Lightweight SQLite relational querying handles multi-user quiz submissions seamlessly.

10. Performance Analysis - Visuals & Re-Test Gains

Interactive Chart.js Dashboard

- **Subject-Wise Bar Chart:** Color-coded visual representation of student performance across all 6 core subjects (Red: <40%, Amber: 40-70%, Green: >70%).
- **Topic-Wise Accuracy Chart:** Horizontal ranking sorted weakest-first to give students an immediate visual roadmap of priorities.
- **Topic Classification Doughnut:** Proportional breakdown displaying percentage of curriculum mastered vs needing revision.
- **Score Progression Line Chart:** Historical trajectory tracking diagnostic scores across successive attempts.

Empirical Re-Test Improvement (ΔScore)

- **Diagnostic Baseline (Attempt 1):** Student S001 scored 52.5% overall; diagnosed with 3 WEAK topics: Backpropagation (25%), Normalization (33%), Word Embeddings (30%).
- **Remedial Intervention:** Student followed the targeted pedagogical pathway: studied curated beginner documentation and worked through step-by-step examples.
- **Post-Remediation (Attempt 2):** Student took the topic re-test and achieved 81.4% overall score.
- **Quantitative Gain:** Demonstrated a positive delta of +28.9 percentage points (ΔScore = +28.9%), validating targeted remediation effectiveness.
- **Pedagogical Outcome:** Weak topics transitioned to 'Strong' (>70%) with zero student disorientation.

11. Applications

Engineering Colleges & Universities

Deployable as a semester diagnostic system 6-8 weeks before university exams to prevent student failure and backlogs.

Faculty Cohort Analytics

Provides instructors with real-time class heatmaps revealing aggregate topic misconceptions, enabling targeted classroom revision lectures.

Outcome-Based Education (OBE)

Offers quantifiable empirical data for Course Outcome (CO) and Program Outcome (PO) direct attainment documentation for NBA accreditation.

Adaptive Self-Study Tool

Empowers students to engage in self-directed deliberate practice rather than passive, inefficient re-reading of textbooks.

Campus Placements & Coding Prep

Extensible to technical interview preparation, aptitude diagnostic assessments, and competitive examinations (e.g. GATE AI&DS).

Institutional Remedial Programs

Automates remedial assignment allocation for slow learners while providing enrichment tasks to advanced learners.

12. Conclusion & Future Scope

Project Conclusions

- **End-to-End System:** Successfully engineered a full-stack, AI-based personalized learning recommendation web platform tailored to B.Tech AI&DS Semester V.
- **Explainable AI Paradigm:** Overcame black-box limitations and cold-start hurdles through a hybrid Decision Tree and Pedagogical Rule Engine.
- **Data-Driven Insights:** Vectorized Pandas analytics automatically triage conceptual weaknesses and present actionable remediation weakest-first.
- **Proven Pedagogical Gains:** Longitudinal score tracking validates substantial improvements (up to +28.9% delta) upon completing recommended learning pathways.

Future Scope & Extensions

- **Item Response Theory (IRT):** Implement computerized adaptive testing where question difficulty dynamically adjusts in real-time based on prior responses.
- **Generative AI Tutoring:** Integrate local Large Language Models (LLMs) to provide interactive Socratic dialogue and dynamic hint generation for weak topics.
- **Multi-Modal Content:** Index timestamped video lectures and interactive Jupyter notebooks directly in the recommendation stream.
- **Cross-Institutional Scaling:** Expand curriculum schemas to cover all 8 semesters across Computer Science, Mechanical, Civil, and Electrical engineering.

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Thanking You!

Any Questions or Suggestions?